Blinkit Sales & Operations Analytics

Quick Commerce Analytics Project — Newton School of Technology

Section E _G-16 Team

NEWTON SCHOOL OF TECHNOLOGY

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The Problem

⏱️ High Delivery Delays

Orders consistently breaching promised delivery windows

😞 Low Satisfaction

Customer ratings and feedback trending negative

🔍 Hidden Inefficiencies

Operational bottlenecks unidentified and unresolved

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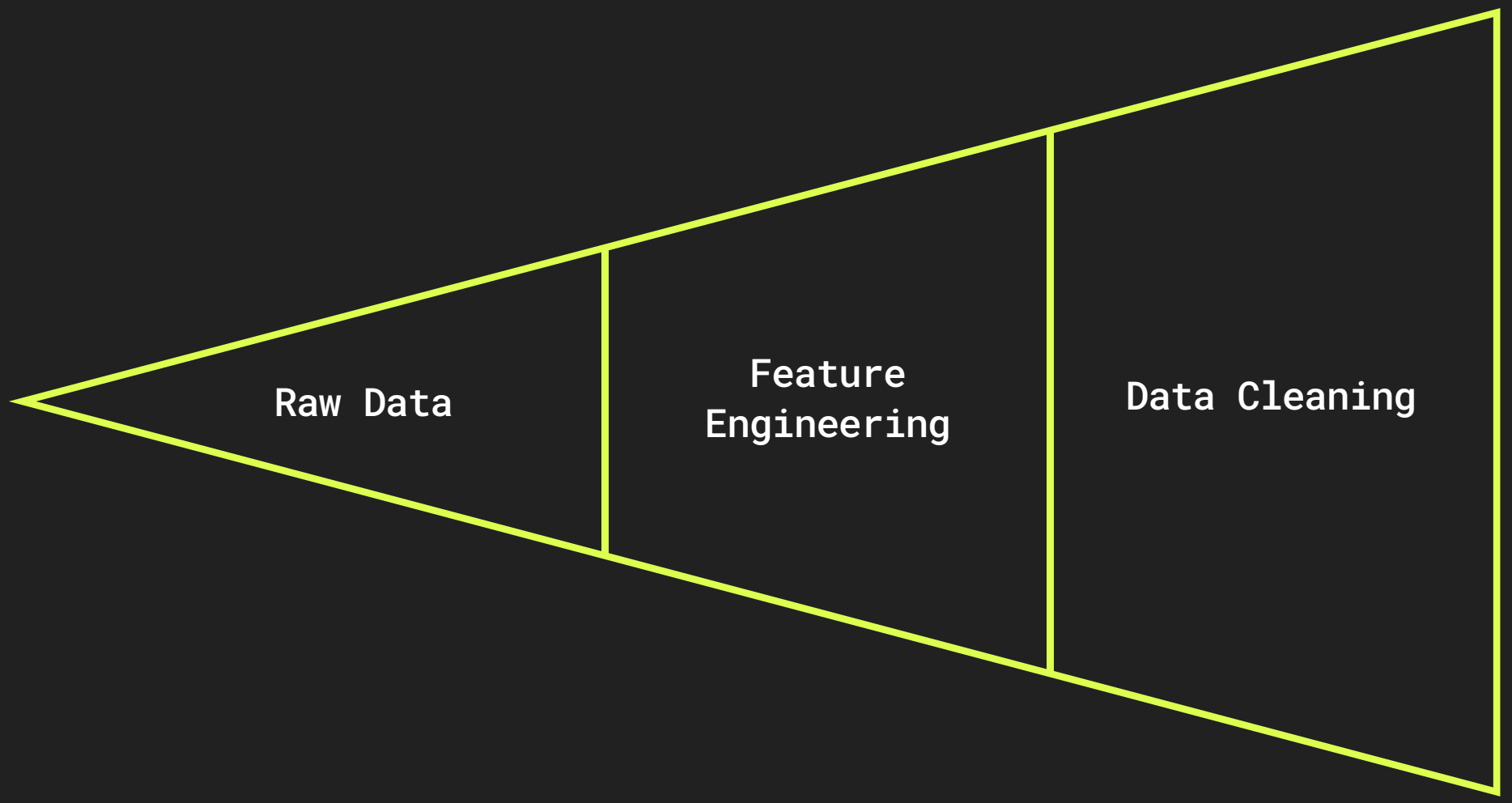
Data Engineering Pipeline



Structured ingestion and transformation ensured analysis-ready data across all operational dimensions.

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KPI Framework

Business

- Revenue
- Orders
- AOV

Operations

- Delay %
- Avg Delay
- Delivery Time

Customer

- Rating
- Repeat %
- Feedback

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EDA Insights

What the Data Shows

- Revenue trend varies significantly by hour and day
- Delay rates spike sharply during peak demand windows
- Certain store clusters consistently underperform

Key Finding

"Operations fail under peak demand" — volume surges expose capacity gaps that standard hours never reveal.

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Statistical Analysis

Methods Applied

- Pearson correlation — delay vs. rating
- Linear regression — delivery time predictors
- Hypothesis testing — peak vs. off-peak performance

Key Result

"Delays negatively impact customer satisfaction" — statistically significant at $p < 0.05$ across all store segments.

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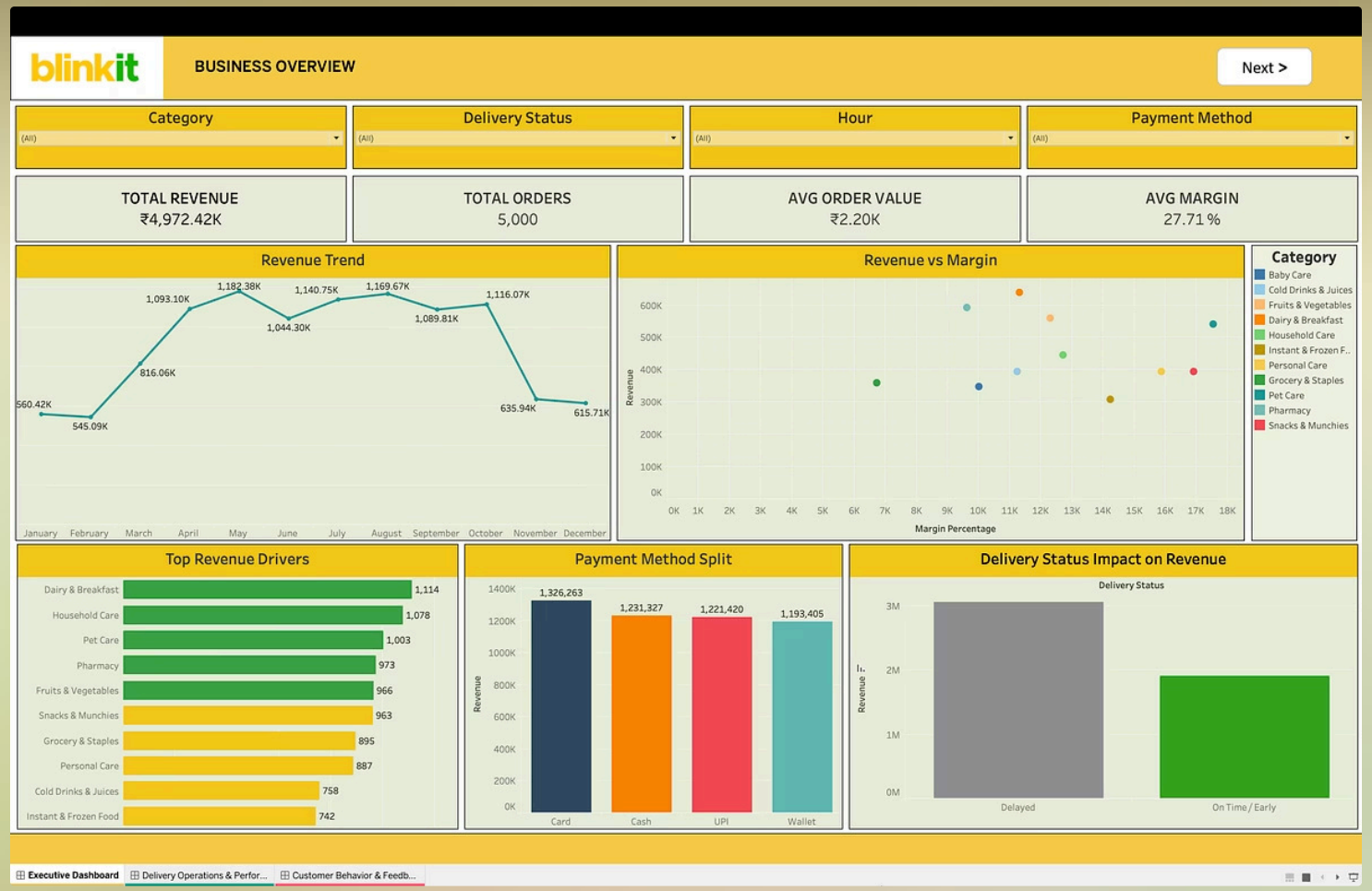

Dashboard – Business Overview

📈 Strong Revenue Performance

Consistent order volumes with healthy AOV trends

📅 Seasonal Variation Observed

Revenue peaks align with promotional periods and weekends

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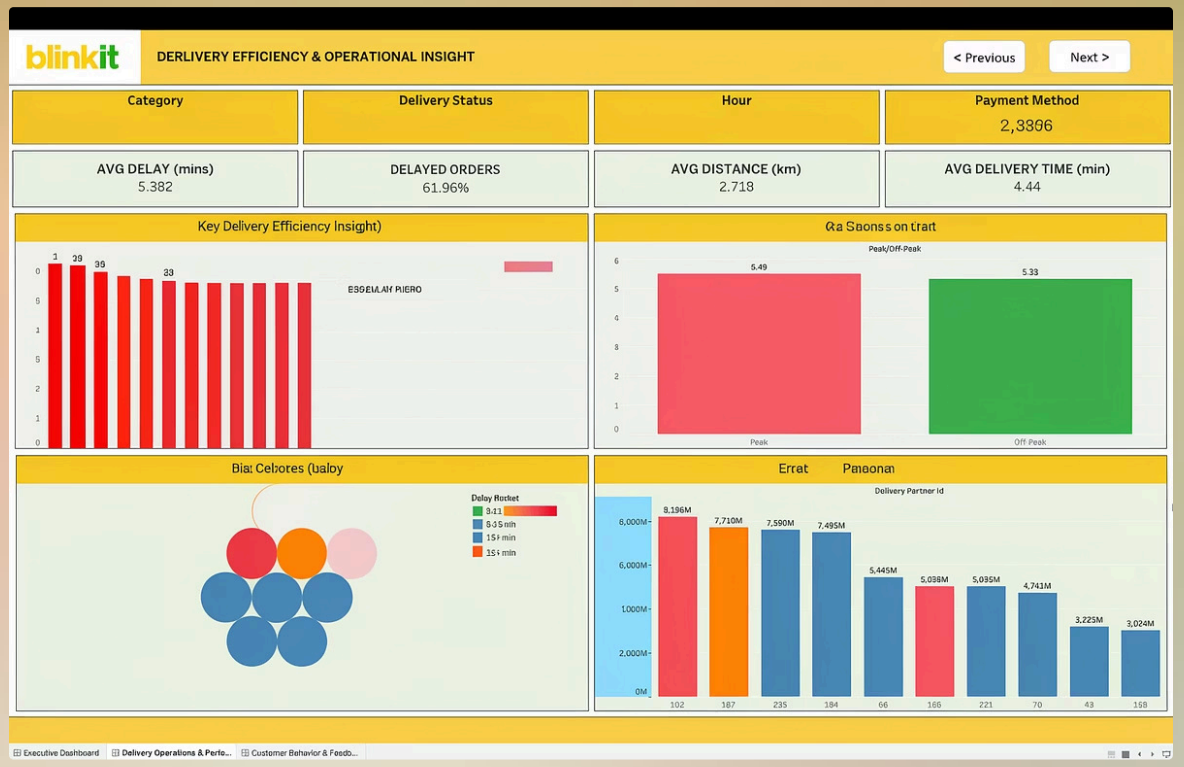
Dashboard – Delivery Operations

🕐 Peak Hours = High Delays

Delay rates double between 6–9 PM across most zones

🏬 Store & Partner Concentration

A small subset of stores and partners cause the majority of issues



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Dashboard – Customer Insights

🔁 High Repeat Rate

Customers return — indicating product-market fit is strong

⚠️ Low Satisfaction

Ratings averaging ~3.3 and feedback skewing negative

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Key Insights & Recommendations

What We Found

- **62%** of orders delayed
- **63%** negative feedback
- Peak hour capacity gap
- Few stores drive most delays
- Avg rating: **3.3 / 5**

What to Do Next

- Scale delivery capacity
- Fix underperforming stores
- Monitor delivery partners
- Improve customer experience

Bottom line: Strong business fundamentals — but improving delivery operations is the clearest path to higher satisfaction and sustainable growth.

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Limitations & Future Scope

Limitations

- ⚠️ Dataset is simulated (not real-time)
- ⚠️ No external factors (traffic, weather)
- ⚠️ Limited historical depth
- ⚠️ Delay calculation assumptions

Future Scope

- Predict delivery delays using ML
- Build real-time dashboards
- Optimize delivery routes
- Customer churn prediction
- Personalized recommendations

While the analysis provides strong insights, integrating real-time data and predictive models can further improve decision-making.

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Impact & Business Value

Delay Reduction

~10–20% improvement in delivery efficiency

Customer Rating Improvement

~0.5–1 point better experience

Retention Increase

5–10% higher repeat customers

Revenue Growth

5–8% due to improved satisfaction

Operational Efficiency

Better store & partner performance

By reducing delays and optimizing operations, the business can improve customer satisfaction, increase retention, and drive revenue growth.

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